



Mark Scheme (Results)

June 2023

Pearson BTEC Nationals
In Applied Science/Forensic and Criminal
Investigation (31617H/1P)
Unit 1: Principles and Applications of Science I
Physics

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Unit 1: Principles and Applications of Science I Physics

General marking guidance

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark grids should be applied positively. Learners must be rewarded for what they have shown they can do rather than be penalised for omissions.
- Examiners should mark according to the mark grid, not according to their perception of where the grade boundaries may lie.
- All marks on the mark grid should be used appropriately.
- All the marks on the mark grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the mark grid.
- Where judgement is required, a mark grid will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the mark grid to a learner's response, a senior examiner should be consulted.

Specific marking guidance

The mark grids have been designed to assess learners' work holistically.

Rows in the grids identify the assessment focus/outcome being targeted. When using a mark grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band depending on how they have evidenced each of the descriptor bullet points.

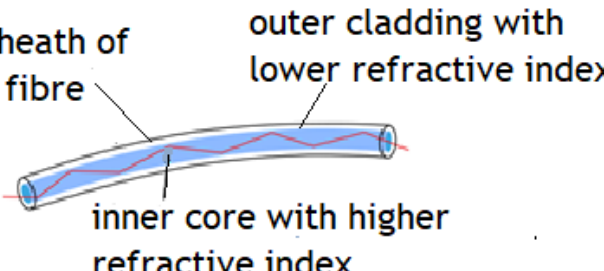
Question Number	Answer	Additional Guidance	Mark
1 (a)(i)	4 (cm)	allow 3.5 (cm) to 4.5(cm) inclusive	1
1 (a)(ii)	0.5(s)	allow 0.4 (s) to 0.6 (s) inclusive accept $\frac{1}{2}$	1
1 (a)(iii)	substitution (1) (f=) $\frac{1}{0.5}$ evaluation (1) 2.0 (Hz)	allow e.c.f. a from 1(a)(ii) allow 2 on its own	2
1 (b)	C destructive, 180°		1
Total			5 marks

Question Number	Answer	Additional Guidance	Mark
2 (a)	compression(s) (1) rarefaction(s) (1)	Answers can be given in either order accept phonetic spellings allow high pressure allow low pressure allow rarefraction, reject rareflection	2
2 (b)	remote controls / optical fibres (1)	accept remotes night/heat vision cameras/mobile phones	1
2 (c)	A description including (the movement is) up and down / side to side/oscillations (1) {at right angles / perpendicular /90°} to the direction{of (travel of)the wave/ of energy (transfer)} (1)		2
Total			5 marks

Question Number	Answer	Additional Guidance	Mark
3 (a)	A description to include any three of the following wave travels {from oscillator/(along/down) the string}(1) wave is reflected/phase change 180°at(the pulley)(at the oscillator)/are not in phase (1) (incident and reflected) waves superimpose/constructive and destructive interference (1) (three) nodes and (four) antinodes are produced (1)	allow bounces back accept energy is stored/no energy transferred/ amplitude is increased/ resonance	3
3 (b)	C third harmonic		1

3 (c)	substitution (1) $32.1 = \sqrt{\frac{I}{6.6 \times 10^{-4}}}$ squaring (1) $1030.41 = \frac{I}{6.6 \times 10^{-4}}$ rearrangement (1) $(I =) 1030.41 \times 6.6 \times 10^{-4}$ evaluation (1) 0.68 (N)	squaring and rearrangement in either order allow 32.1^2 award 3 marks for 6.8 to any other power of ten 0.021 gains 3 marks (not squared) 2 marks for 2.1 to any other power of ten answers between 4.4×10^{-4} and 4.5×10^{-4} gains 3 marks (squared at end of calculation) answers between 4.4 and 4.5 to any other power of ten gains 2 marks 0.82 gains 3 marks 8.2 to any other power of ten gains 2 marks 0.146/0.145 gains 2 marks 1.46/1.45 to any other power of ten gains 1 mark If no mark scored award 1 mark for 32.1×6.6 with any power of ten seen in calculation	4
Total			8 marks

Question Number	Answer	Additional Guidance	Mark
4 (a)	D radio wave, microwave, infrared		1
4 (b)	An explanation linking the following for either type of phone described Terrestrial Mobile phone (mobile phone) transmits a (coded) signal (radio waves/microwaves) to the (nearest) base station / phone mast/tower/hub (1) {base station / phone mast/tower/hub} transmit signal to (central exchange / main computer / server} (1) Idea of the signal travelling from (base station / phone mast/tower/hub) to another mobile phone (1) OR Satellite mobile phone (satellite) phone transmits a signal (radio waves/microwaves/upload) to the satellite(1) the frequency/wavelength of the signal is changed (1) the signal is downloaded and received by another satellite phone/ base station / mobile phone.(1)	Allow change in frequency/wavelength Allow this to be the same phone mast that the signal was sent to	3
4 (c)	Bluetooth® has a {limited/short} range/ (1) Bluetooth® devices are very low powered / approximately 1mW/ weak(er) signal (1)	allow cannot travel far allow 10m range ignore poor signal allow for two marks 'there is a strong(er) signal at a short(er) distance'/ weak signal unless the distance is short	2
Total			6 marks

Question Number	Indicative content
5	Structure of endoscope • two bundles of optical fibres • optical fibres are very thin fibres of glass about the thickness of a hair • each optical fibre has cladding • bundles are inside a flexible tube • light passes down one bundle (incoherent bundle) • to light the area being observed • light is reflected from the inside of the stomach • image is passed up through the second (coherent) bundle • image of inside of the stomach passed up to computer screen • each optical fibre of the second bundle gives one pixel of the image seen on the screen Passage of light through fibre optics • light passes down the optical fibre by total internal reflection • the light strikes the side of the optical fibre • the incident angle is greater than the critical angle between the fibre and the cladding • the light is reflected • the cladding increases the critical angle so that there are less reflections down the length of the fibre • less energy is lost as there are less reflections 

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates adequate knowledge of scientific facts / concepts with generalised comments made. • Generic statements may be presented rather than linkages made so that lines of reasoning are unsupported or partially supported. • The explanation shows some structure and coherence.
Level 2	3–4	• Demonstrates good knowledge and understanding by selecting and applying some relevant scientific facts / concepts to provide the discussion being presented. • Lines of argument mostly supported through the application of relevant evidence. • The explanation shows a structure which is mostly clear, coherent and logical
Level 3	5–6	• Demonstrates comprehensive knowledge and understanding by selecting and applying relevant knowledge of scientific facts / concepts to provide the discussion being presented. • Line(s) of argument consistently supported throughout by sustained application of relevant evidence. • The explanation shows a well-developed structure which is clear logical and coherent.
Total 6 marks		



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